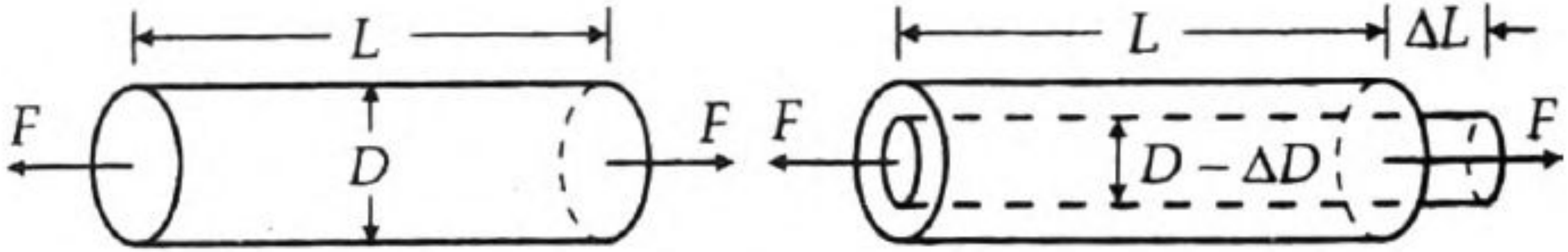
STRAIN GAUGES

If a metal conductor is stretched or compressed, its resistance changes on account of the fact that both length and diameter of conductor change. Also there is a change in the value of resistivity of the conductor when it is strained and this property is called piezoresistive effect. Therefore, resistance strain gauges are also known as piezoresistive gauges. The strain gauges are used for measurement of strain and associated stress in experimental stress analysis.

Secondly, many other detectors and transducers, notably the load cells, torque meters, diaphragm type pressure gauges, temperature sensors, accelerometers and flow meters, employ strain gauges as secondary transducers.

Theory of Strain Gauges

The change in the value of resistance by straining the gauge may be partly explained by the normal dimensional behaviour of elastic material. If a strip of elastic material is subjected to tension, as shown in Fig. or in other words positively strained, its longitudinal dimension will increase while there will be a reduction in the lateral dimension. So when a gauge is subjected to a positive strain, its length increases while its area of cross-section decreases as shown in Fig . Since the resistance of a conductor is proportional to its length and inversely proportional to its area of cross-section, the resistance of the gauge increases with positive strain. The change in the value resistance of strained conductor is more than what can be accounted for an increase in resistance due to dimensional changes. The extra change in the value of resistance is attributed to a change in the value of resistivity of a conductor when strained. This property, as described earlier, is known as peizoresistive effect.



Change in dimensions of a strain gauge element when subjected to a tensile force.

Let us consider a strain gauge made of circular wire. The wire has the dimensions : length = L, area = A, diameter = D before being strained. The material of the wire has a resistivity ρ.

Resistance of unstrained gauge R = ρL/ A.

Let a tensile stress s be applied to the wire. This produces a positive strain causing the length to increase and area to decrease.

Thus when the wire is strained there are changes in its dimensions. Let ΔL = change in length, ΔA = change in area, ΔD = change in diameter and ΔR = change in resistance.

In order to find how ΔR depends upon the material physical quantities, the expression for R is differentiated with respect to stress s.

Thus we get :

$$\frac{dR}{ds} = \frac{\rho}{A}\frac{\partial L}{\partial s} - \frac{\rho L}{A^2}\frac{\rho A}{\partial s} + \frac{L}{A}\frac{\partial \rho}{\partial s} \qquad \text{...(25.59)}$$

Dividing Eqn. 25.59 throughout by resistance $R = \rho L / A$, we have

$$\frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} - \frac{1}{A}\frac{\partial A}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \qquad \text{...(25.60)}$$

It is evident from Eqn. 25.60, that the per unit change in resistance is due to :

(i) per unit change in length - $\Delta L / L$

(ii) per unit change in area = $\Delta A / A$ and

(iii) per unit change in resistivity -$\Delta \rho / \rho$

$$\text{Area} \qquad A = \frac{\pi}{4} D^2 \quad \therefore \quad \frac{\partial A}{\partial s} = 2.\frac{\pi}{4} D.\frac{\partial D}{ds} \qquad \text{...(25.61)}$$

$$\text{or} \qquad \frac{1}{A}\frac{dA}{ds} = \frac{(2\pi/4)D}{(\pi/4)D^2}\frac{\partial D}{\partial s} = \frac{2}{D}\frac{\partial D}{\partial s} \qquad \text{...(25.62)}$$

$\therefore$ Equation 25.60 can be written as :

$$\frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} - \frac{2}{D}\frac{\partial D}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \qquad \text{...(25.63)}$$

Now, Poisson's ratio

$$\nu = \frac{\text{lateral strain}}{\text{longitudinal strain}} = -\frac{\partial D / D}{\partial L / L} \qquad \ldots(25.64)$$

or $$\partial D / D = -\nu \times \partial L / L$$

$$\therefore \quad \frac{1}{R}\frac{dR}{ds} = \frac{1}{L}\frac{\partial L}{\partial s} + \nu\frac{2}{L}\frac{\partial L}{\partial s} + \frac{1}{\rho}\frac{\partial \rho}{\partial s} \qquad \ldots(25.65)$$

For small variations, the above relationship can be written as :

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + 2\nu\frac{\Delta L}{L} + \frac{\Delta \rho}{\rho} \qquad \ldots(25.66)$$

Gauge factor

The gauge factor is defined as the ratio of per unit change in resistance to per unit change in length.

Gauge factor $$G_f = \frac{\Delta R / R}{\Delta L / L} \quad \ldots(25.67)$$

or $$\frac{\Delta R}{R} = G_f \frac{\Delta L}{L} = G_f \times \varepsilon \qquad \text{...(25.68)}$$

where $\varepsilon = \text{strain} = \dfrac{\Delta L}{L}$

The gauge factor can be written as :

$$= 1 + 2\nu + \frac{\Delta\rho/\rho}{\varepsilon} \qquad \text{...(25.69)}$$

$$= \underbrace{1}_{\substack{\text{Resistance}\\ \text{change due to}\\ \text{change of length}}} + \underbrace{2\nu}_{\substack{\text{Resistance}\\ \text{change due to}\\ \text{change in area}}} + \underbrace{\frac{\Delta\rho/\rho}{\varepsilon}}_{\substack{\text{Resistance}\\ \text{change due to}\\ \text{piezoresistive effect}}}$$

$$G_f = \frac{\Delta R/R}{\Delta L/L} = 1 + 2\nu + \frac{\Delta\rho/\rho}{\Delta L/L}$$

The strain is usually expressed in terms of microstrain. 1 microstrain = $1\,\mu m / m$.

If the change in the value of resistivity of a material when strained is neglected, the gauge factor is :

$$G_f = 1 + 2\nu \qquad \text{...(25.70)}$$

Equation 25.70 is valid only when **Piezoresistive Effect** *i.e.,* change in resistivity due to strain is almost negligible.

The Poisson's ratio for all metals is between 0 and 0.5. This gives a gauge factor of approximately, 2. The common value for Poisson's ratio for wires is 0.3. This gives a value of 1.6 for wire wound strain gauges.

Table 25.4 *Gauge Factors*

Material	***Gauge***	***Material***	***Gauge Factor***
Nickel	– 12.1	Platinum	+ 4.8
Manganin	+ 0.47	Carbon	+ 20
Nichrome	+ 2.0	Doped Crystals	100 – 5000
Constantan	+ 2.1		
Soft iron	+ 4.2		

Applications

Strain gauges are broadly used for two major types of applications and they are :

(i) Experimental stress analysis of machines and structures, and

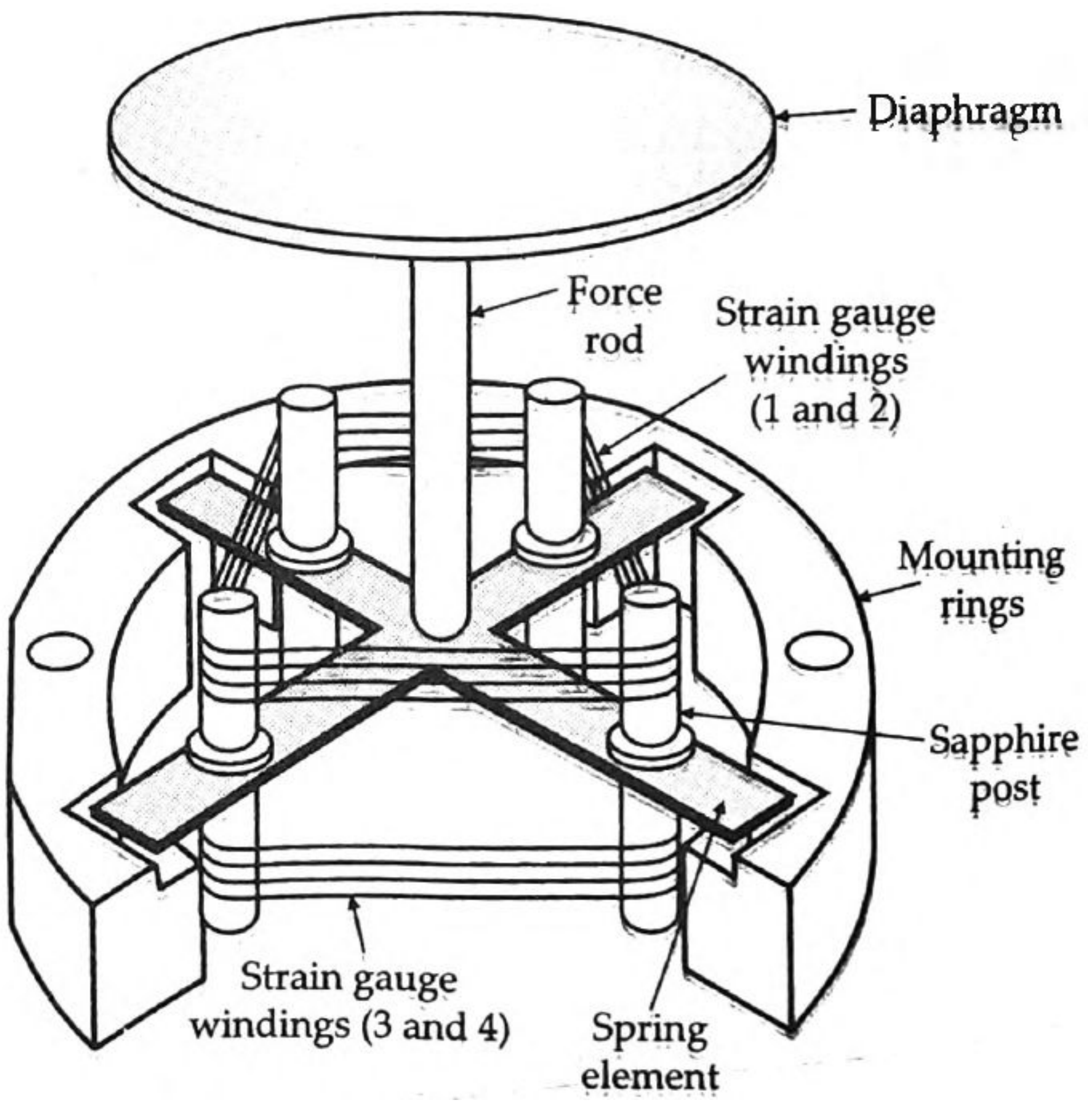
(ii) Construction of force, torque, pressure, flow and acceleration transducers.

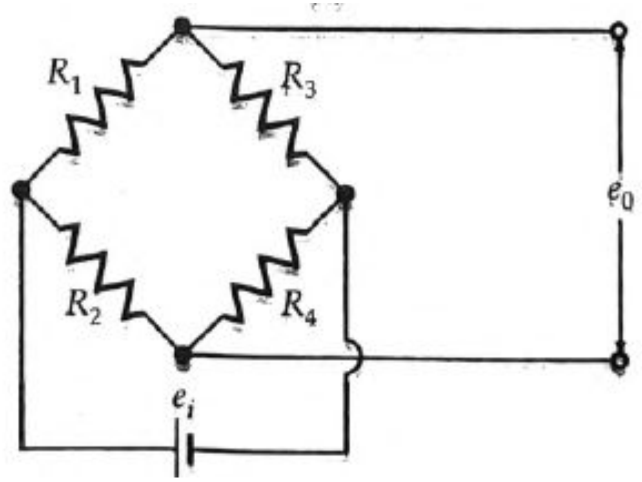
TYPES OF STRAIN GAUGES

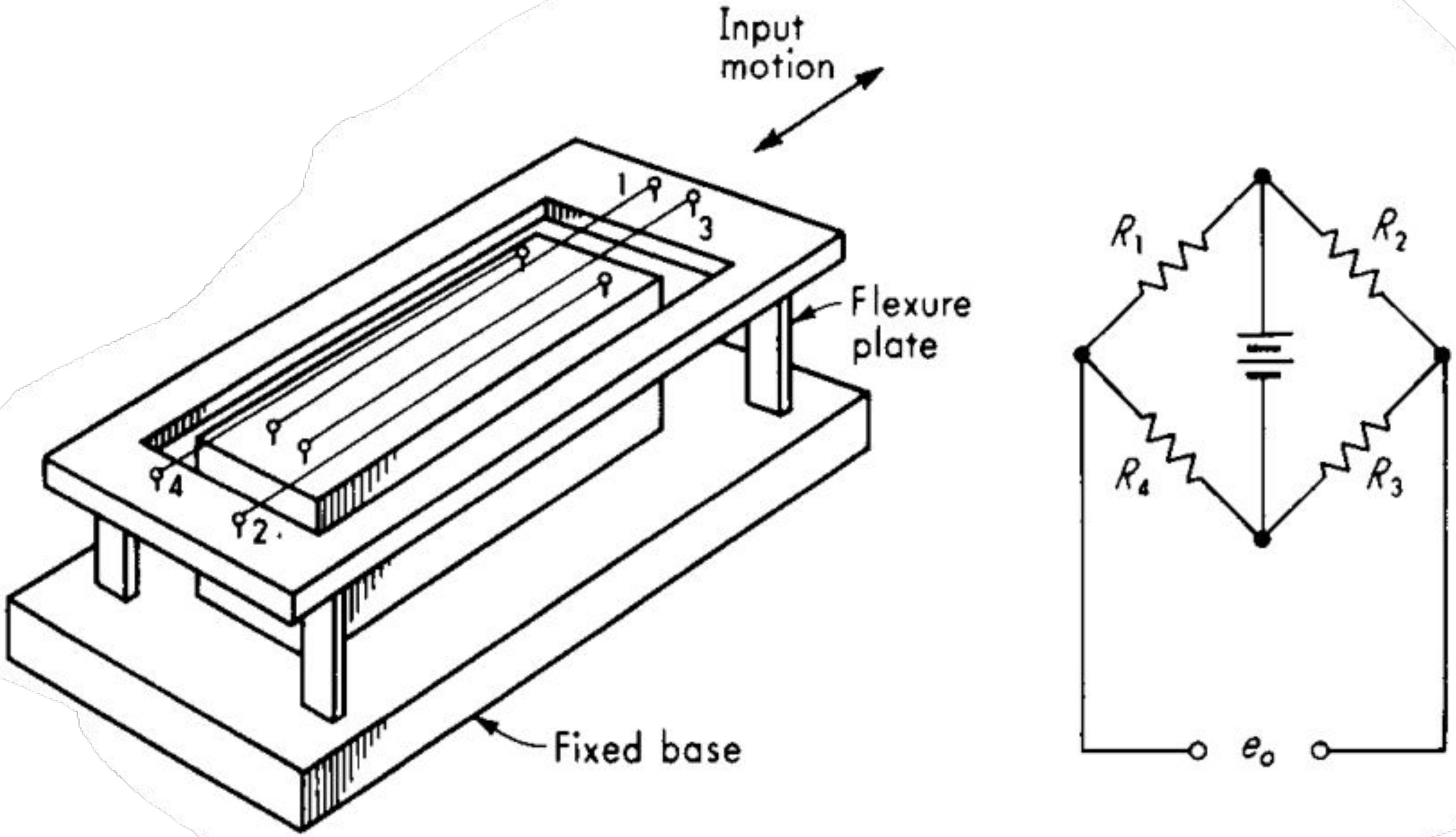
1. Unbonded metal strain gauges
2. Bonded metal wire strain gauges
3. Bonded metal foil strain gauges
4. Vacuum deposited thin metal film strain gauges
5. Sputter deposited thin metal strain gauges
6. Bonded semiconductor strain gauges
7. Diffused metal strain gauges.

Unbonded metal strain gauges

This gauge consists of a wire stretched between two points in an insulating medium such as air. The wires may be made of various copper nickel, chrome nickel or nickel iron alloys. They are about 0.003 mm in diameter, have a gauge factor of 2 to 4 and sustain force of 2 mN. The length of wire is 25 mm or less.


Diaphragm
Force rod
Strain gauge windings (1 and 2)
Mounting rings
Sapphire post
Strain gauge windings (3 and 4)
Spring element


R_1
R_3
e_0
R_2
R_4
e_i

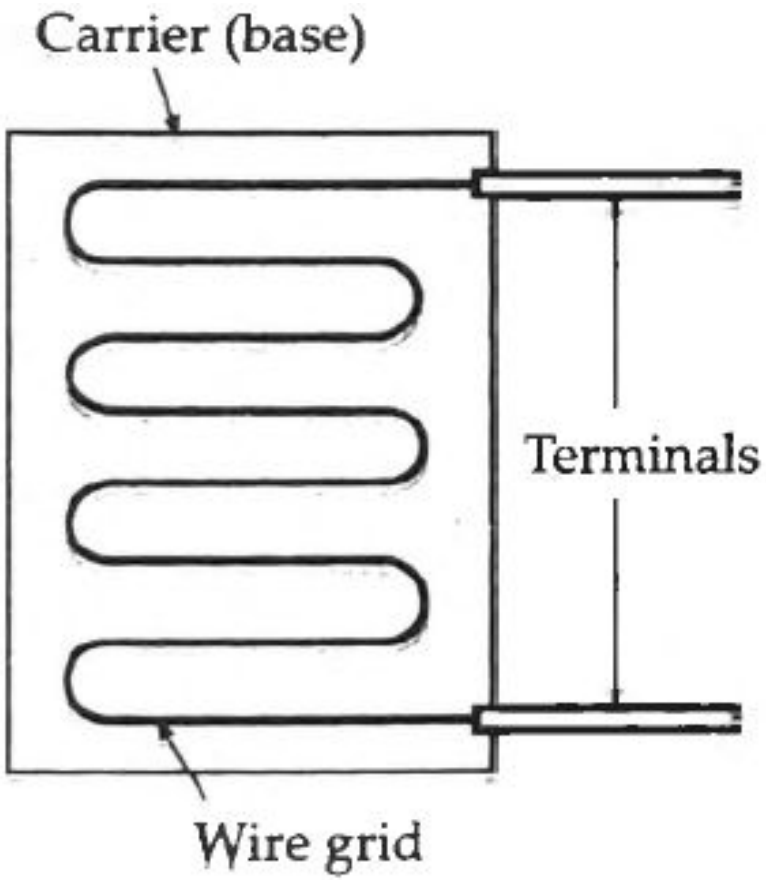


Input motion
1
3
4
2
Flexure plate
Fixed base
R_1
R_2
R_4
R_3
e_o

The flexture element is connected via a rod to a diaphragm which is used for sensing of pressure. The wires are tensioned to avoid buckling when they experience a compressive force.

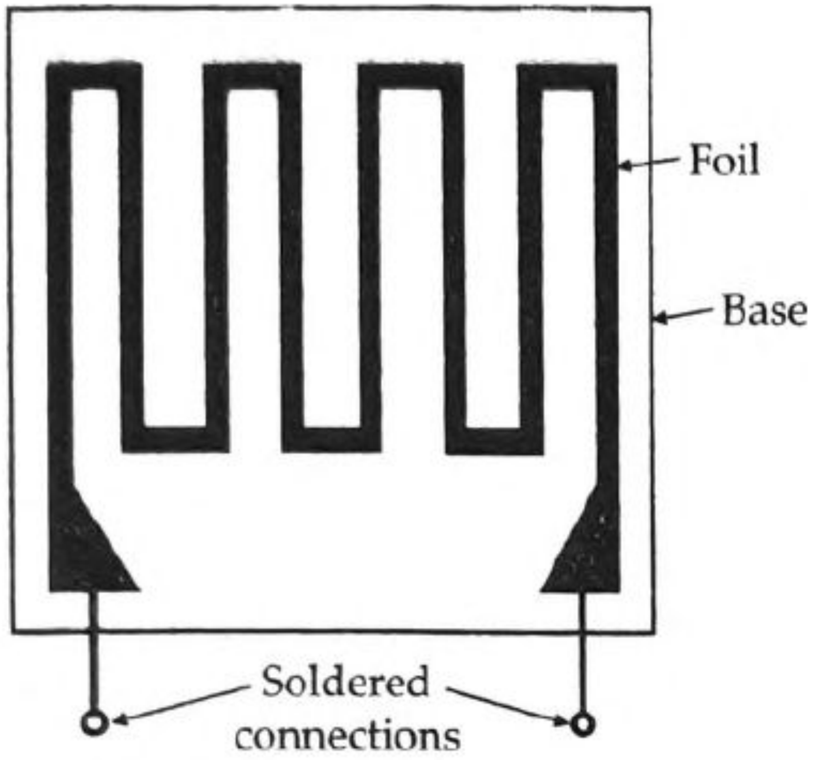
Bonded Wire Strain Gauges

The bonded metal-wire strain gauges are used for both stress analysis and for construction of transducers.A resistance wire strain gauge consists of a grid of fine resistance wire of about 0.025 mm in diameter or less. The grid is cemented to carrier (base) which may be a thin sheet of paper, a thin sheet of bakelite or a sheet of teflon. The wire is covered on top with a thin sheet of material so as to prevent it from any mechanical damage. The spreading of wire permits a uniform distribution of stress over the grid. The carrier is bonded with an adhesive material to the specimen under study. This permits a good transfer of strain from carrier to grid of wires. The wires cannot buckle as they are embedded in a matrix of cement and hence faithfully follow both the tensile and compressive strains of the specimen.

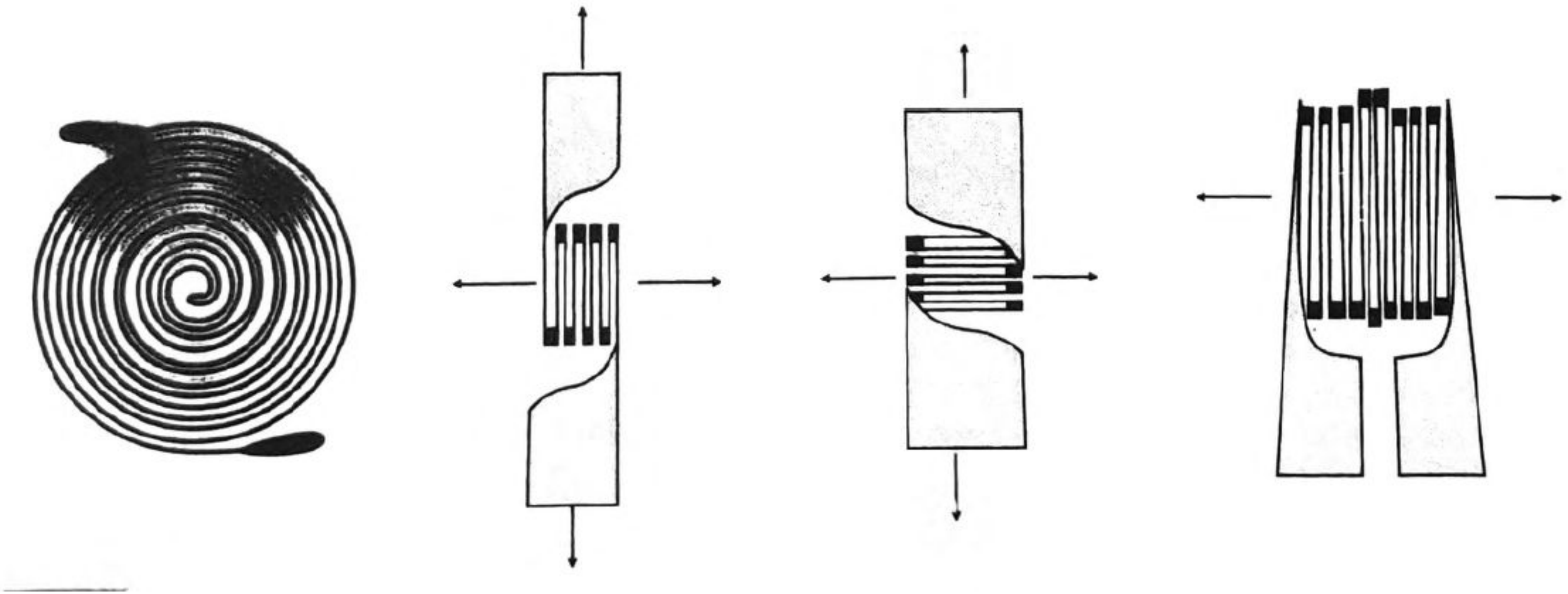

Carrier (base)
Terminals
Wire grid

Bonded Metal Foil Strain Gauges

This class of strain gauges is only an extension of the bonded metal wire strain gauges. The bonded metal wire strain gauges have been completely superseded by bonded metal foil strain gauges.Metal foil strain gauges use identical or similar materials to wire strain gauges and are used today for most general-purpose stress analysis applications and for many transducers.

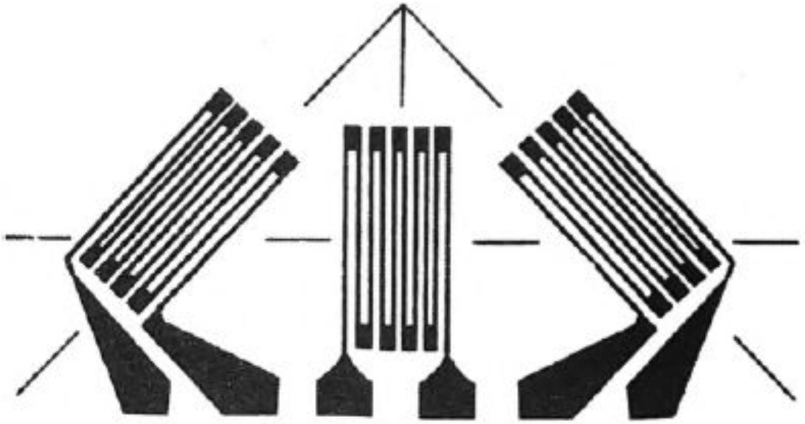

Foil
Base
Soldered
connections

The sensing elements of foil gauges

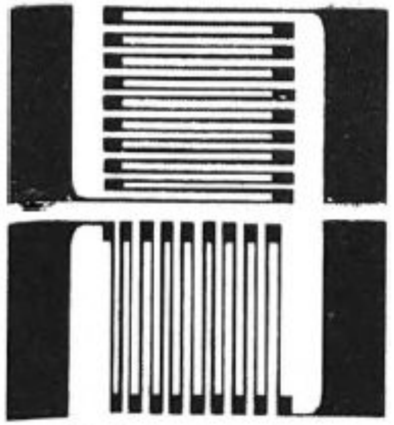


ROSETTES

In addition to single element strain gauges, a combination of strain gauges called "Rosettes" are available in many combinations for specific stress analysis or transducer applications.



FAER-25RB-12SX
3-element rosette
45° planar
(Foil)



2-element rosette
90° planar
(Foil)

FAED-25B-35SX
2-element rosette
90° shear planar
(Foil)